

.ekbf File Format Specification

Version 1 — Architecture & Implementation Reference

1 Global File Layout

.ekbf (Emoji Kitchen Binary Format) is a compact binary layout optimized for zero-allocation memory-mapped reads ($O(1)$ direct lookups) and dense storage of lower-triangular relationship matrices.

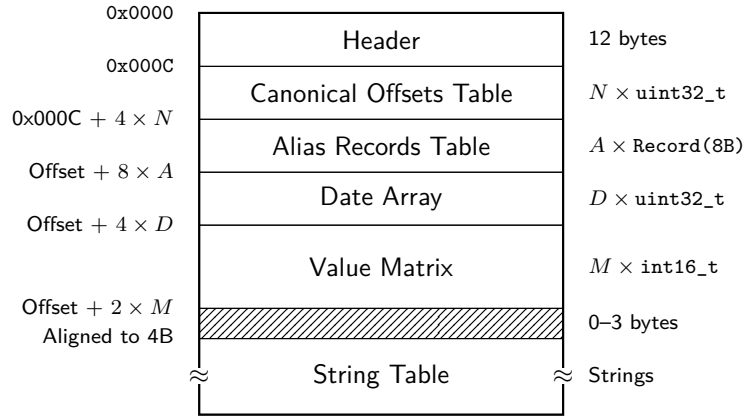


Figure 1: .ekbf Top-Level Memory Layout Structure

1.1 Overview & Section Definitions

The file consists of a fixed 12-byte Header followed by six contiguous payload sections. All multibyte integer primitives are stored in **Little-Endian** byte ordering. The String Table section is guaranteed to begin on a **4-byte alignment boundary**; if necessary, 1 to 3 null padding bytes (0x00) follow the Value Matrix.

Section	Element Type	Total Size	Description
Header	Composite	12 Bytes	Fixed magic identifier ("EKBF") and metadata counters.
Canonical Offsets	<code>uint32_t</code>	$4 \times N$ Bytes	Offsets into String Table for canonical entity names.
Alias Records	Struct (8B)	$8 \times A$ Bytes	ID mapping schema pairing aliases to target canonical IDs.
Date Array	<code>uint32_t</code>	$4 \times D$ Bytes	Numerical dates (YYYYMMDD), one per unique release date.
Value Matrix	<code>int16_t</code>	$2 \times M$ Bytes	Packed lower triangular data matrix ($M = \frac{N(N+1)}{2}$).
Alignment Padding	Byte array	0 to 3 Bytes	Null-byte padding (0x00) aligning String Table to 4B.
String Table	C-Strings	Variable	Null-terminated (\0) UTF-8 encoded string block.

Table 1: Summary of .ekbf File Sections

2 Section Specifications

2.1 Header Layout

The file header is exactly 12 bytes. It defines the magic identifier ("EKBF"), format version, and element counts (N, A, D) needed to compute every later section's offset without scanning the file.

0x00	Magic Number	4 Bytes ([4]byte)
0x04	Version	2 Bytes (uint16_t)
0x06	Canonical (N)	2 Bytes (uint16_t)
0x08	Alias (A)	2 Bytes (uint16_t)
0x0A	Dates (D)	2 Bytes (uint16_t)
0x0C		

Figure 2: Header Structure Offset Diagram (12 Bytes Total)

Offset	Field	Type	Size	Description
0x00	Magic Number	[4]byte	4 Bytes	Fixed ASCII signature: ['E', 'K', 'B', 'F'].
0x04	Version	uint16_t	2 Bytes	Specification version (current: 1).
0x06	Canonical (N)	uint16_t	2 Bytes	Count of canonical entities.
0x08	Alias (A)	uint16_t	2 Bytes	Count of alias records.
0x0A	Dates (D)	uint16_t	2 Bytes	Count of unique dates. Max 0x7FFF; see §2.5.2.

Table 2: Detailed Header Field Specifications

2.2 Canonical Offsets Table

The Canonical Offsets Table is an array of N contiguous `uint32_t` values starting immediately after the Header at byte offset 0x000C. Each element holds an absolute byte offset pointing into the String Table where the entity's null-terminated hex string begins. An entity's index in this array ($0 \dots N - 1$) is its *canonical ID*.

2.3 Alias Records Table

The Alias Records Table contains A consecutive 8-byte structures. Each record maps an alias entity string to its target canonical ID.

Struct Offset	Field	Type	Size	Description
0x00	Alias Offset	uint32_t	4 Bytes	Offset into String Table for the alias hex string.
0x04	Target Canonical ID	uint32_t	4 Bytes	Index into the Canonical Offsets Table.

Table 3: Alias Record Structure Schema (8 Bytes per entry)

2.4 Date Array

The Date Array contains D values encoded as 32-bit unsigned integers (`uint32_t`), each a calendar date formatted as YYYYMMDD (e.g., 20260726 for July 26, 2026). Dates are addressed by a 0-based index; matrix cells reference an entry in this array via the encoding described in §2.5.2, not by direct byte offset.

2.5 Value Matrix

To maximize space efficiency, the format stores only the **lower triangular** elements of the $N \times N$ relationship matrix, including the diagonal. The matrix occupies $M = \frac{N(N+1)}{2}$ signed 16-bit integers (`int16_t`).

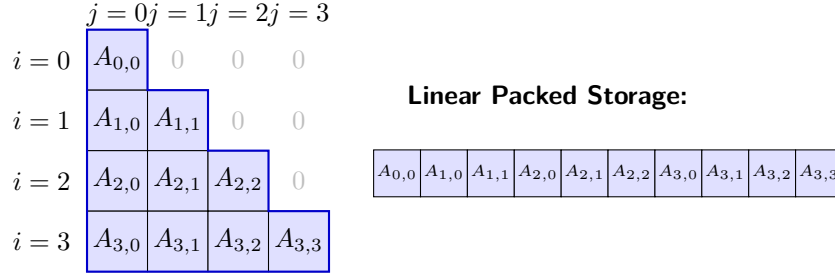


Figure 3: Lower Triangular Matrix Mapping to 1D Array (illustrative 4×4 example; the real dimension is N)

2.5.1 Lower Triangular Index Computation

For a matrix with N dimensions, any entry $A_{i,j}$ (where $i \geq j$) maps to its linear array index $\text{pos}(i, j)$:

$$\text{pos}(i, j) = \frac{i(i+1)}{2} + j$$

If $i < j$, swap operands: $\text{pos}(j, i)$.

2.5.2 Cell Value Encoding

A cell value of 0 indicates no known combination exists. A non-zero cell value packs two things into one signed `int16_t`:

- **Magnitude** ($|\text{value}|$): one plus the 0-based index into the Date Array, i.e. $\text{dateIndex} = |\text{value}| - 1$.
- **Sign**: orientation of the resulting asset. A *positive* value means row operand i is primary; a *negative* value means column operand j is primary.

Because one bit of range is spent on sign rather than magnitude, the largest usable magnitude is 32767; the value -32768 is never written, as it has no positive counterpart. This caps the format at 32767 unique dates.

2.6 String Table

The String Table contains sequential null-terminated (`0x00`) UTF-8 encoded strings. Offsets into this table are relative to the start of the String Table section, i.e. offset 0 is the section's first byte.